

$$u \cdot w =$$

$$w_i \cdot p_{j+1} = w_i \left((v_{j+1} \cdot w_i) \cdot w_i \right)$$

$$w_i (v_{j+1} - p_{j+1})$$

$$w_i v_{j+1} - w_i \cdot (v_{j+1} \cdot w_i) \cdot w_i$$

$$\sum (v+w)^2 = \sum (v-w)^2$$

$$\lambda = a + bi$$

$$A u + A w_i = \lambda u + \lambda w_i$$

$$\mathbb{R} \ni v \cdot \bar{v} \geq 0$$

$$\begin{aligned} \lambda v \cdot \bar{v} &= A v \cdot \bar{v} = v^T A^T \bar{v} \\ &= v^T \cdot A \bar{v} \end{aligned}$$

$$v \in \mathbb{C}^n$$

$$v = u + w i \quad u, w \in \mathbb{R}^n$$

$$A \bar{v} = A v - 2 A w i$$

$$= \lambda v - 2 \lambda w i$$

$$\lambda$$

